

pyproject.toml

A `pyproject.toml` file is the standard configuration file for modern Python projects. It tells Python tools how to build, install, format, test, and manage your project.

It was introduced by Python Software Foundation through PEP standards like PEP 518 and PEP 621.

A `pyproject.toml` can define:

- Project metadata (name, version, author)
- Dependencies
- Build system configuration
- Tool settings (formatter, linter, test runner, etc.)

It replaces or consolidates older files like:

- `setup.py`
- `setup.cfg`
- `requirements.txt` (sometimes)
- tool-specific config files

Example

```
[project]
name = "my_app"
version = "0.1.0"
description = "Example Python project"
dependencies = [
    "requests",
    "fastapi"
]

[build-system]
requires = ["setuptools>=61"]
build-backend = "setuptools.build_meta"
```

Common sections

[project]

Defines package metadata.

```
[project]
name = "cool-library"
version = "1.2.3"
```

[build-system]

Tells Python how to build/install the package.

```
[build-system]
requires = ["setuptools", "wheel"]
build-backend = "setuptools.build_meta"
```

[tool.*]

Configuration for developer tools.

Example:

```
[tool.black]
line-length = 88

[tool.pytest.ini_options]
testpaths = ["tests"]
```

Why it's called TOML

The file format is TOML (“Tom’s Obvious, Minimal Language”), designed to be easy for humans to read.

Example syntax:

```
key = "value"
number = 123
list = ["a", "b"]
```

Typical project structure

```
my_project/
├─ pyproject.toml
├─ src/
├─ tests/
└─ README.md
```